

Using FFT in an Oscilloscope

Technical Report

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Chapter 1

Introduction

In engineering education, students study Fourier analysis as a mathematical tool to understand signals in both time and frequency domains. However, many students find it difficult to relate this theory to real-world applications.

This report explains how the **Fast Fourier Transform (FFT)** is practically used in an oscilloscope to analyze real electrical systems, such as a refrigerator compressor. The aim is to clearly bridge theoretical concepts taught in university courses with practical diagnostics carried out using modern instruments.

Chapter 2

Why FFT Is Used Instead of Fourier Transform

2.1 Fourier Transform in Theory

In theory, students learn several types of Fourier transforms:

- Continuous-Time Fourier Transform (FT)
- Discrete-Time Fourier Transform (DTFT)
- Discrete Fourier Transform (DFT)

The continuous Fourier Transform is defined as:

$$X(f) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi ft} dt \quad (2.1)$$

This equation requires infinite data over infinite time, which makes it impossible to implement directly in real hardware.

2.2 Limitations of FT and DTFT

An oscilloscope cannot compute:

- FT – because it requires infinite integration
- DTFT – because it requires infinite summation

In real systems, only a finite number of samples can be captured.

2.3 FFT as the Practical Solution

The oscilloscope uses the **Fast Fourier Transform (FFT)**, which is an efficient algorithm to compute the DFT:

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j2\pi kn/N} \quad (2.2)$$

Key Point:

FFT is not a different transform; it is simply a fast way of calculating the DFT.

Chapter 3

Why Frequency Domain Analysis Is Necessary

3.1 Limitations of Time-Domain Analysis

Time-domain waveforms show only the signal shape. A compressor current waveform may appear as a clean sine wave, even when hidden problems exist.

3.2 Hidden Information in Frequency Domain

Using FFT, the following can be revealed:

- Harmonics
- Electrical noise
- PWM switching components
- Mechanical vibration frequencies
- Resonance effects

These components are invisible in the time domain but clearly visible in the frequency spectrum.

Chapter 4

Connecting Engineering Theory to Practical Measurement

4.1 Analog Nature of Compressor Current

The refrigerator compressor draws alternating current, which is an analog signal similar to those studied in Signals and Systems courses.

4.2 Sampling and ADC

The signal is sampled using an ADC:

$$x[n] = x(nT_s) \quad (4.1)$$

This directly corresponds to sampling theory taught in DSP.

4.3 Time-Domain Representation

The current can be modeled as:

$$i(t) = I_{\max} \sin(2\pi 50t) + \text{distortion} + \text{noise} \quad (4.2)$$

4.4 FFT Output

The FFT output represents:

$$|X(f)| \quad (4.3)$$

which shows the signal's frequency components.

Chapter 5

Harmonics, Noise, and Nonlinearity

5.1 Harmonics Due to Nonlinearity

Nonlinear loads generate harmonics. In a compressor motor, FFT typically shows:

- 100 Hz (2nd harmonic)
- 150 Hz (3rd harmonic)
- 200 Hz (4th harmonic)

This directly validates harmonic analysis theory.

5.2 High-Frequency Noise

Switching elements such as MOSFETs and SMPS circuits generate high-frequency noise. FFT reveals these components as sharp spikes at higher frequencies.

Chapter 6

Mechanical Vibration Analysis Using FFT

FFT is also used to detect mechanical faults:

- Low-frequency vibration (10–60 Hz)
- Rotor imbalance
- Resonance frequencies

This technique is widely used in condition monitoring and predictive maintenance.

Chapter 7

Why FFT Is the Preferred Practical Tool

FFT is widely used because:

- It works with finite sampled data
- It is computationally efficient
- It allows real-time analysis
- It can be implemented on oscilloscopes and microcontrollers

FFT is used in:

- Power quality analysis
- Motor diagnostics
- EMI/EMC testing
- Audio and communication systems

Chapter 8

FFT-Based Fault Diagnosis of Compressors

Fault	FFT Signature
Weak capacitor	Strong 150 Hz harmonic
Faulty relay	High startup harmonics
Mechanical vibration	Low-frequency peaks
Inverter failure	Missing switching frequency
Overload	Excess low-frequency energy

Table 8.1: Compressor Faults Identified Using FFT

Chapter 9

Simple Intuitive Explanation of FFT

FFT works like separating sound into individual musical notes. Although a signal looks simple in time domain, it contains multiple frequency components that FFT separates clearly.

Chapter 10

Conclusion

This report demonstrated how FFT bridges the gap between theoretical Fourier analysis and real-world engineering applications. The oscilloscope-based FFT analysis used for compressor diagnostics is a direct implementation of concepts taught in Signals and Systems and DSP courses.

FFT enables engineers to detect faults, analyze noise, and understand system behavior efficiently, making it one of the most important tools in modern engineering practice.